PLSC 30600 — Homework 3

Due: Fri, February 27 (11:59pm)

General instructions

- **Show your work** for all hand calculations. A final numeric answer without intermediate steps, formulas, and/or justification is not sufficient.
- For coding problems, submit a reproducible script (`.R/.rmd/.rnw`) along with the fully compiled pdf. Set a random seed and report it.
- Include a brief note describing how you used AI tools (if at all), consistent with the course AI policy.

Problem 1: Binary IV and the LATE theorem

In this problem, you will work through the binary-instrument special case of the Imbens–Angrist LATE theorem. Throughout, assume $Z_i \in \{0,1\}$, $D_i \in \{0,1\}$, and define potential receipt $D_i(1), D_i(0)$ and potential outcomes $Y_i(1), Y_i(0)$.

IV assumptions (as in the slides).

- **SUTVA:** $D_i = D_i(1)Z_i + D_i(0)(1 - Z_i)$ and $Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$.
- **Instrument independence:** $\big(Y_i(0), Y_i(1), D_i(1), D_i(0)\big) \perp Z_i$.
- **Relevance:** $\mathrm{E}[D_i \mid Z_i = 1] \neq \mathrm{E}[D_i \mid Z_i = 0]$.
- **Exclusion:** outcomes depend on receipt D only (i.e., $Y_i(d, z) = Y_i(d)$; no direct effect of Z on Y).

Important: do *not* invoke monotonicity yet; allow for defiers.

1.a Start with the first stage. Define the (population) intent-to-treat effect on receipt:

$$ITT_D \equiv \mathrm{E}[D_i \mid Z_i = 1] - \mathrm{E}[D_i \mid Z_i = 0].$$

Using SUTVA and instrument independence, show that

$$ITT_D = \mathrm{E}[D_i(1) - D_i(0)]\,.$$

1.b Show that ITT_D is "compliers minus defiers" (no monotonicity). Define compliance types by the pair $\big(D_i(0), D_i(1)\big)$. Prove that

$$ITT_D = \Pr[\text{complier}] - \Pr[\text{defier}].$$

Hint: write $D_i(1) - D_i(0)$ as a function of type.

1.c Interpretation (short answer). Explain why this result makes relevance ($ITT_D \neq 0$) an assumption about the presence/absence of compliers and defiers.

1.d Now do the reduced form (numerator). Define the (population) intent-to-treat effect on outcomes:

$$ITT_Y \equiv \mathrm{E}[Y_i \mid Z_i = 1] - \mathrm{E}[Y_i \mid Z_i = 0].$$

Using SUTVA, instrument independence, and exclusion, show that

$$ITT_Y = \mathrm{E}\big[Y_i\big(D_i(1)\big) - Y_i\big(D_i(0)\big)\big]\,.$$

1.e Show that ITT_Y is a type-weighted difference (no monotonicity). Using the compliance types defined above and the exclusion restriction, prove that

$$ITT_Y = \Pr[\text{complier}] \cdot \mathrm{E}[Y(1) - Y(0) \mid \text{complier}] - \Pr[\text{defier}] \cdot \mathrm{E}[Y(1) - Y(0) \mid \text{defier}]\,.$$

1.f Put the numerator and denominator together (still no monotonicity). Define the Wald estimand (binary Z):

$$\tau_{Wald} \equiv \frac{ITT_Y}{ITT_D}.$$

Using your answers above, show that

$$\tau_{Wald} = \frac{\Pr[\text{complier}] \cdot \mathrm{E}[Y(1) - Y(0) \mid \text{complier}] - \Pr[\text{defier}] \cdot \mathrm{E}[Y(1) - Y(0) \mid \text{defier}]}{\Pr[\text{complier}] - \Pr[\text{defier}]}.$$

1.g Wald without monotonicity (short answer + algebra). Define $LATE_C \equiv \mathrm{E}[Y(1) - Y(0) \mid \text{complier}]$ and $ATE_F \equiv \mathrm{E}[Y(1) - Y(0) \mid \text{defier}]$ to simplify the expression.

Show that you can rewrite τ_{Wald} as

$$\tau_{Wald} = w_C\, LATE_C + w_F\, ATE_F,$$

where $w_C + w_F = 1$ but $w_F < 0$ whenever $\Pr[\text{defier}] > 0$. Give explicit formulas for w_C and w_F, and briefly interpret why the weight on defiers is negative.

1.h Now impose monotonicity and prove the LATE theorem (binary case). Add the monotonicity assumption (no defiers):

$$D_i(1) \geq D_i(0) \quad \text{for all } i.$$

Conclude that $\Pr[\text{defier}] = 0$, so that $ITT_D = \Pr[\text{complier}]$, and show that

$$\tau_{Wald} = \frac{ITT_Y}{ITT_D} = \mathrm{E}[Y(1) - Y(0) \mid D(1) - D(0) = 1]\,.$$

Problem 2: Fuzzy RD is IV (a local Wald ratio)

One way to connect IV to RD is that **fuzzy RD** is an IV design: crossing a cutoff induces a jump in treatment probability, but does not deterministically assign treatment.

Let R_i be a running variable with cutoff c. Define the cutoff indicator

$$Z_i \equiv \mathbf{1}\{R_i \geq c\},$$

let D_i be actual treatment receipt (which may be imperfectly determined by Z_i), and Y_i the outcome.

2.a Sharp vs. fuzzy RD estimands. Define the discontinuity in conditional expectations at the cutoff:

$$\Delta_Y \equiv \lim_{r \downarrow c} \mathrm{E}[Y \mid R = r] - \lim_{r \uparrow c} \mathrm{E}[Y \mid R = r], \qquad \Delta_D \equiv \lim_{r \downarrow c} \mathrm{E}[D \mid R = r] - \lim_{r \uparrow c} \mathrm{E}[D \mid R = r].$$

What is the **sharp RD** estimand in terms of Δ_Y when $D = Z$ (deterministic assignment at the cutoff)? What is the **fuzzy RD** (Wald) estimand in terms of Δ_Y and Δ_D?

2.b Connect to IV. Explain (2–5 sentences) why the fuzzy RD estimand is a Wald ratio with instrument $Z = \mathbf{1}\{R \geq c\}$, but where the comparison is made "locally" around $R = c$. Who are the "compliers" in the fuzzy RD setting?

2.c Assumptions (RD language ↔ LATE language). List the main assumptions needed for the fuzzy RD Wald ratio to have a LATE interpretation at the cutoff, and map each one to the corresponding LATE/IV assumption (independence/as-good-as-random, exclusion, monotonicity, relevance).

Coding A: FWL partialling out for IV (simulated data)

This coding problem shows how the "partialling out" (Frisch–Waugh–Lovell) idea works for IV. You will compute three versions of the IV estimate:

(i) the raw Wald ratio (no covariates),

(ii) the covariate-adjusted Wald ratio using FWL residualization, and

(iii) the "canned" 2SLS estimate from software.

Simulated dataset. Use the following data-generating process:

```
set.seed(30600)
n <- 3000
X <- rnorm(n)
U <- rnorm(n)

# Instrument assignment depends on X (so Z is NOT unconditionally randomized),
# but is independent of U given X.
Z <- rbinom(n, 1, plogis(0.5 * X))

# Endogenous receipt D: depends on Z, X, and unobserved U
v <- rnorm(n)
D_star <- -0.2 + 0.8 * Z + 0.7 * X + 1.0 * U + v
D <- as.numeric(D_star > 0)

# Outcome depends on D, X, and U
tau <- 1.5
eps <- rnorm(n)
Y <- tau * D + 0.5 * X + 1.0 * U + eps

df <- data.frame(Y, D, Z, X)
```

A.a Inspect the simulated data. Report n, $\bar{D}$, and the first-stage difference in means $\bar{D}_{Z=1} - \bar{D}_{Z=0}$. Also report $\bar{X}_{Z=1} - \bar{X}_{Z=0}$. (It should not be exactly zero.)

A.b IV by hand (no covariates): Wald ratio. Compute

$$\widehat{ITT}_Y = \bar{Y}_{Z=1} - \bar{Y}_{Z=0}, \qquad \widehat{ITT}_D = \bar{D}_{Z=1} - \bar{D}_{Z=0}, \qquad \hat{\tau}_{Wald} = \frac{\widehat{ITT}_Y}{\widehat{ITT}_D}.$$

Report $\hat{\tau}_{Wald}$.

A.c IV with covariates using FWL residualization. Residualize Y, D, and Z on X using OLS:

$$\tilde{Y}_i \equiv Y_i - \widehat{\mathrm{E}}[Y_i \mid X_i], \qquad \tilde{D}_i \equiv D_i - \widehat{\mathrm{E}}[D_i \mid X_i], \qquad \tilde{Z}_i \equiv Z_i - \widehat{\mathrm{E}}[Z_i \mid X_i].$$

Concretely, this means:

(i) run three separate regressions: Y on X, D on X, and Z on X (each with an intercept);

(ii) keep the residuals from each regression, which are the parts of Y, D, Z not explained by X.

(If X were a *vector* of controls, you would regress on all columns of X at once; the idea is the same.)

Then compute the **covariate-adjusted Wald/IV** estimate:

$$\hat{\tau}_{FWL} = \frac{\sum_{i=1}^{n} \tilde{Z}_i \tilde{Y}_i}{\sum_{i=1}^{n} \tilde{Z}_i \tilde{D}_i}.$$

Report $\hat{\tau}_{FWL}$. (It should be much closer to $\tau = 1.5$ than the raw Wald ratio.)

A.d Canned 2SLS using software. Use `AER::ivreg()` to estimate 2SLS with the same covariate X included in both stages:

$$Y \sim D + X \quad \text{instrumenting } D \text{ with } Z\text{, and treating } X \text{ as its own instrument.}$$

Report the coefficient on D and its standard error. Confirm it matches $\hat{\tau}_{FWL}$ up to small numerical tolerance.

Coding B: Replicate Angrist–Krueger (1991)

In this question you will replicate parts of Angrist and Krueger (1991), "Does compulsory school attendance affect schooling and earnings?'' (QJE). The key idea is to instrument years of education with quarter of birth.

Reference: Angrist, Joshua D. and Alan B. Krueger. 1991. "Does compulsory school attendance affect schooling and earnings?'' *The Quarterly Journal of Economics* 106(4): 979–1014.

The dataset is available at: https://raw.githubusercontent.com/UChicago-pol-methods/PLSC30600/refs/heads/main/data/angrist-krueger.csv

The variables are:

- `log_weekly_wage`: log weekly wage (outcome)
- `education`: years of education (endogenous regressor)
- `quarter_of_birth`: $Z \in \{1, 2, 3, 4\}$ (instrument)
- `year_of_birth`: year of birth (coded as integers)
- `place_of_birth`: place of birth (coded as integers)

B.a Read and inspect. Read in the data, report n, and report the set of unique values of `quarter_of_birth` and `year_of_birth`.

B.b Table III, Panel B. Re-generate Table III, Panel B. You may use `lm` and `AER::ivreg()` for standard errors.

B.c Table V (many instruments). Re-generate the years-of-education coefficient and standard error from only the first two columns of Table V. Report only the `education` coefficient and its standard error (no need to report all dummy coefficients, test statistics, etc.).

B.d Brief interpretation. Compare your Table III (Panel B) 2SLS estimate to your Table V estimate.

B.e LATE theorem conditions: are they plausible here? (short essay) List the conditions required for the LATE theorem interpretation of IV/2SLS (in the Imbens–Angrist framework):

- SUTVA / well-defined potential outcomes
- independence (as-good-as-random instrument, possibly conditional on controls)
- exclusion (no direct effect of the instrument on the outcome)
- monotonicity (no defiers, relative to the instrument shift)
- relevance (nontrivial first stage)

For each condition, briefly discuss whether it seems satisfied in the Angrist–Krueger quarter-of-birth instrument setting, and why (or why not). Be concrete about what the "compliers" are in this application and what causal effect your 2SLS estimate can be interpreted as.